FINAL-2026A
STATISTICAL COMPUTING

Name: ______________________________ Roll Number: ____________________

TIME: 2 HRS
MAXIMUM MARKS: 60

This paper has nine questions. Question 1 is a subjective question and you have to show all your work to get full marks. It carries 12 marks. If you do this problem at the beginning of your answer script, you will get extra 2 points. Questions 2-9 are short answer type questions and there is no partial marking. All carry 6 marks. You have to just write down the answers in this question paper in the designated places. No work needs to be shown. We will not check any rough work. Do not do any rough work in the question paper, you will loose two points. Write your name and roll numbers at the top of each page, otherwise two marks will be deducted. Submit this question paper along with the answer booklet. We will be using the following notations: The probability density function (PDF), $X \sim \text{Gamma}(\alpha, \lambda, \mu)$, for $\alpha > 0$, $\lambda > 0$, $-\infty < \mu < \infty$, means X has PDF $f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)}(x-\mu)^{\alpha-1}e^{-\lambda(x-\mu)}$, for $x > \mu$. $X \sim \text{Weibull}(\alpha, \lambda)$ for $\alpha > 0, \lambda > 0$, means X has PDF $f(x) = \alpha\lambda x^{\alpha-1}e^{-\lambda x^\alpha}$, for $x > 0$. $X \sim \text{Exp}(\lambda)$, for $\lambda > 0$, means X has PDF $f(x) = \lambda e^{-\lambda x}$, for $x > 0$. $X \sim \text{Poisson}(\lambda)$ means $P(X = i) = \frac{e^{-\lambda}\lambda^i}{i!}$, for $i = 0, 1, \ldots$. It is assumed that you know how to generate samples only from an uniform distribution.

1. Let $\phi(x)$ and $\Phi(x)$ be the PDF and cumulative distribution function (CDF) of a standard normal random variable, respectively. Consider the following function for $-\infty < x < \infty$ and $\lambda > 0$:

$$F(x) = \lambda \int_0^\infty \Phi\left(\frac{x-u}{u}\right) e^{-\lambda u} du.$$

(a) Show that $F(x)$ satisfies all the properties of a proper distribution function and find PDF of $F(x)$ in terms of $\phi(x)$.
(b) Give an explicit method to generate samples from the CDF $F(x)$ when $\lambda = 2$.
[$4 + 8 = 12$]

2. Suppose $\{X_1, \ldots, X_n\}$ is a random sample from $\text{Gamma}(\alpha, 1, \mu)$. Let $A = \frac{1}{n}\sum_{i=1}^n x_i$ and $B = \frac{1}{n}\sum_{i=1}^n x_i^2$. If $A = 5$, $B = 27$ and $\tilde{\mu}$ and $\tilde{\alpha}$ the method of moment estimators of μ and α, respectively, then

Answer
(a) $\tilde{\mu} =$

(b) $\tilde{\alpha} =$

3. Suppose $\{X_1, \ldots, X_n\}$ is a random sample from Gamma(α, λ). Here both α and λ are unknown. Let $A = \sum_{i=1}^{n} \ln x_i$ and $B = \sum_{i=1}^{n} x_i$. Further, let $\hat{\alpha}$ and $\hat{\lambda}$ denote the MLEs of α and λ, respectively. Suppose, the MLE of α can be obtained by maximizing a function $g(\alpha)$. If $n = 15$, $A = 10$, $B = 12$, then

Answer
(i) $g(\alpha) =$

(ii) $\frac{\hat{\alpha}}{\hat{\lambda}} =$

4. Suppose $U_1 \sim$ Exp(2) and $U_2 \sim$ Exp(3) are they are independently distributed. If $A = E(U_1 | U_1 > 1)$ and $B = E(U_1 | \min\{U_1, U_2\} = 2)$, then

Answer
(i) $A =$

(ii) $B =$

5. The total number of accidents during the i-th day of a week in a particular busy junction follows Poisson distribution with parameter λ_i; $i = 1, \ldots, 7$. It is assumed that $\lambda_i = \alpha + i\beta$, and we have the following observations for seven days $\{1, 1, 1, 1, 1, 1, 1\}$. We want to compute the MLEs of α and β. Suppose when $\beta = 1$, we have to solve the non-linear equation $g_1(\alpha) = 0$ to compute the MLE of α and when $\alpha = 0$, $\hat{\beta}$ is the MLE of β. Then

Answer:
(a) $g_1(\alpha) =$

(b) $\hat{\beta} =$

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6. Suppose $f(x)$, the PDF of a random variable X, has the following form: $f(x) = pf_1(x) + (1-p)f_2(x)$, where $f_1(x)$ and $f_2(x)$ are also PDFs and $0 < p < 1$. Suppose, $f_1(x) = 2\lambda|x|e^{-\lambda|x|^2}$, if $x < 0$ and 0, otherwise. $f_2(x) = 4\lambda x e^{-2\lambda x^2}$ for $x > 0$ and 0, otherwise. You have the following sample $\{-4, -3, -2, -1, 1, 2, 3\}$. Let $\hat{p}$ and $\hat{\lambda}$ be the MLEs of p and λ, respectively. Then

Answer:
(a) $\hat{p} =$

(b) $\hat{\lambda} =$

7. Suppose $\{X_1, \ldots, X_n\}$ is a random sample from the following PDF

$$f(x;\theta) = 1 \quad \text{if} \quad \theta - \frac{1}{2} < x < \theta + \frac{1}{2}$$

and 0, otherwise. Let $X_{(1)} = \min\{X_1, \ldots, X_n\}$ and $X_{(n)} = \max\{X_1, \ldots, X_n\}$. We want to find the maximum product of spacings (MPS) estimator of θ. Suppose $x_{(1)} = 9.0$ and $x_{(n)} = 9.5$, and Suppose $g(\theta)$ is a function which needs to be maximized with respect to θ to compute the MPS estimator of θ. Let $\tilde{\theta}$ denote the MPS estimator of θ. Then

Answer:
(a) $g(\theta) =$

(b) $\tilde{\theta} =$

8. Let $U_1 \sim$ Weibull$(2, 2)$, $U_2 \sim$ Weibull$(2, 3)$ and $U_3 \sim$ Weibull$(2, 4)$ and they are independently distributed. Let $X = \min\{U_1, U_3\}$ and $Y = \min\{U_2, U_3\}$ and $f_{X,Y}(x, y)$ denote the joint PDF of X and Y. If $f_1(x, y) = f_{X,Y}(x, y)$ for $x < y$ and $f_2(x, y) = f_{X,Y}(x, y)$, for $x > y$, then

Answer:
(a) $f_1(x, y) =$

(b) $f_2(x, y) =$

9. Suppose $f(x)$, the PDF of a random variable X, has the following form for $\alpha \geq 0$, $\beta \geq 0$ and $\alpha + \beta > 0$:

$$f(x; \alpha, \beta) = \begin{cases} (\alpha + 2\beta x)e^{-\alpha x - \beta x^2} & \text{if} \quad x > 0 \\ 0 & \text{if} \quad x \leq 0 \end{cases}$$

Let $\delta = \frac{\alpha}{\beta}$ if $\beta \neq 0$. We have a random sample $\{t_1, \ldots, t_{10}\}$ from the above $f(x)$, and we know $10 = \sum_{i=1}^{10} t_i$, $10 = \sum_{i=1}^{10} t_i^2$, $7 = \sum_{i=1}^{10} \ln(1 + t_i)$ and $4 = \sum_{i=1}^{10} \ln(1 + 0.5t_i)$. It is known that the possible values of (α, δ) are $\{(1,1), (1,2), (2,1), (2,2)\}$ only. We want to compute the MLEs of α and δ. If $\delta = 1$, then MLE of α is denoted by $\tilde{\alpha}(1)$. If both are unknown, the MLEs of (α, δ) are denoted by $(\tilde{\alpha}, \tilde{\delta})$. You can use the fact that $2 < e$. Then

Answer:
(a) $\tilde{\alpha}(1) =$

(b) $\tilde{\alpha} + 3\tilde{\delta} =$

Wishing you all the best